Yaswanth Sreenivas D

240701611

RAD (Rapid Application Development) is a software development methodology that emphasizes **quick development and iteration** of prototypes over strict planning and long development cycles. It's ideal for projects where requirements evolve through user feedback and where fast delivery is important.

Key Characteristics of RAD:

- Focus on **prototypes** instead of extensive documentation
- **User involvement** throughout the development process
- **Fast development cycles** with frequent iterations
- **Reusable components** to reduce effort
- Best suited for **UI/UX-intensive projects** like web and mobile apps

<u>Phase 1</u>: Requirements Planning

This is where the **project goals, system requirements, and user needs** are identified. Stakeholders, users, and designers collaborate to decide what needs to be built.

1) Identify Key Features

Navigation:

- Home
- Product Details
- Product Categories
- Cart
- Product Listings
- Order Confirmation

User Actions: Browsing, Searching, Adding to Cart, Checkout, Tracking Orders

2) Create a Requirements Document:

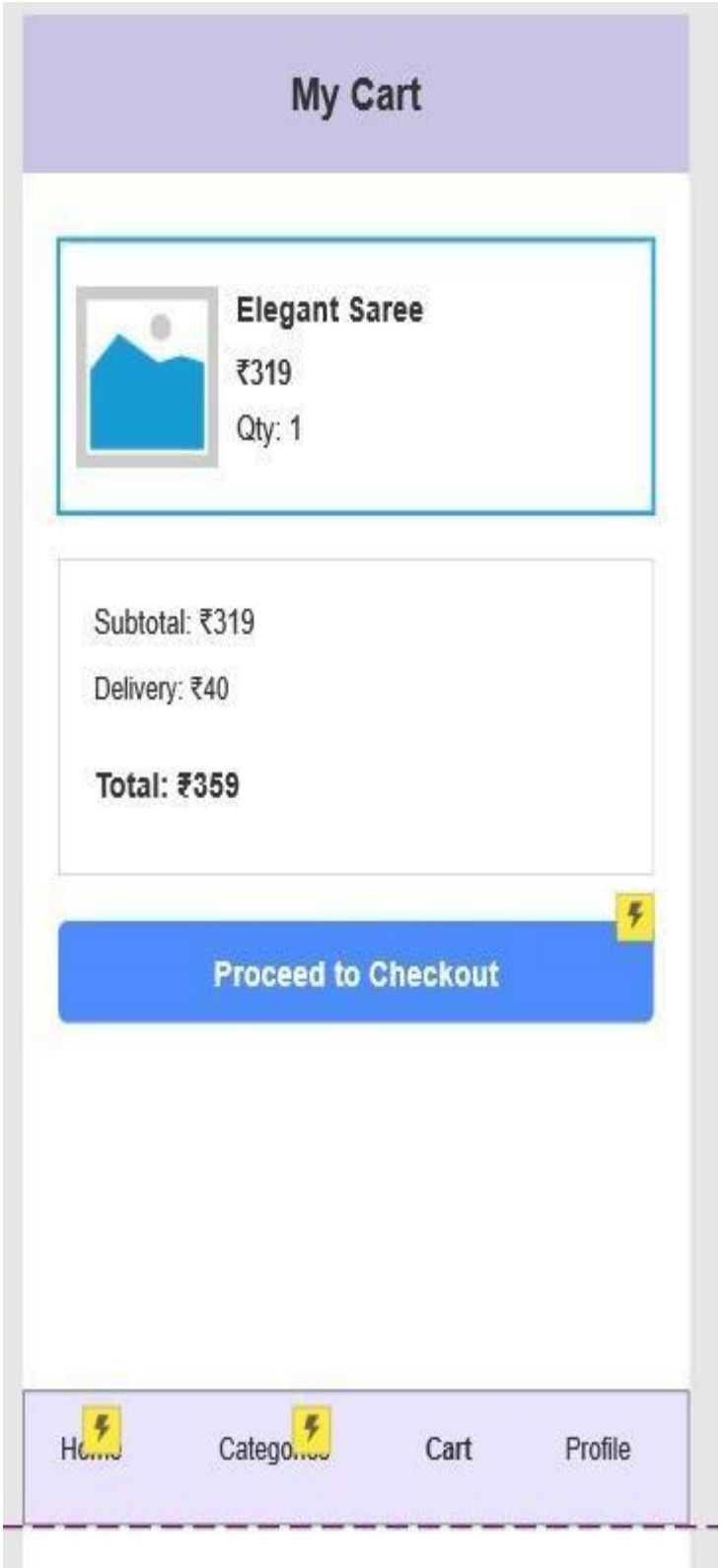
1. View list of product categories
2. View products under each category
3. View detailed information about a product
4. View and edit cart
5. Checkout and enter user/shipping/payment details
6. View order confirmation message

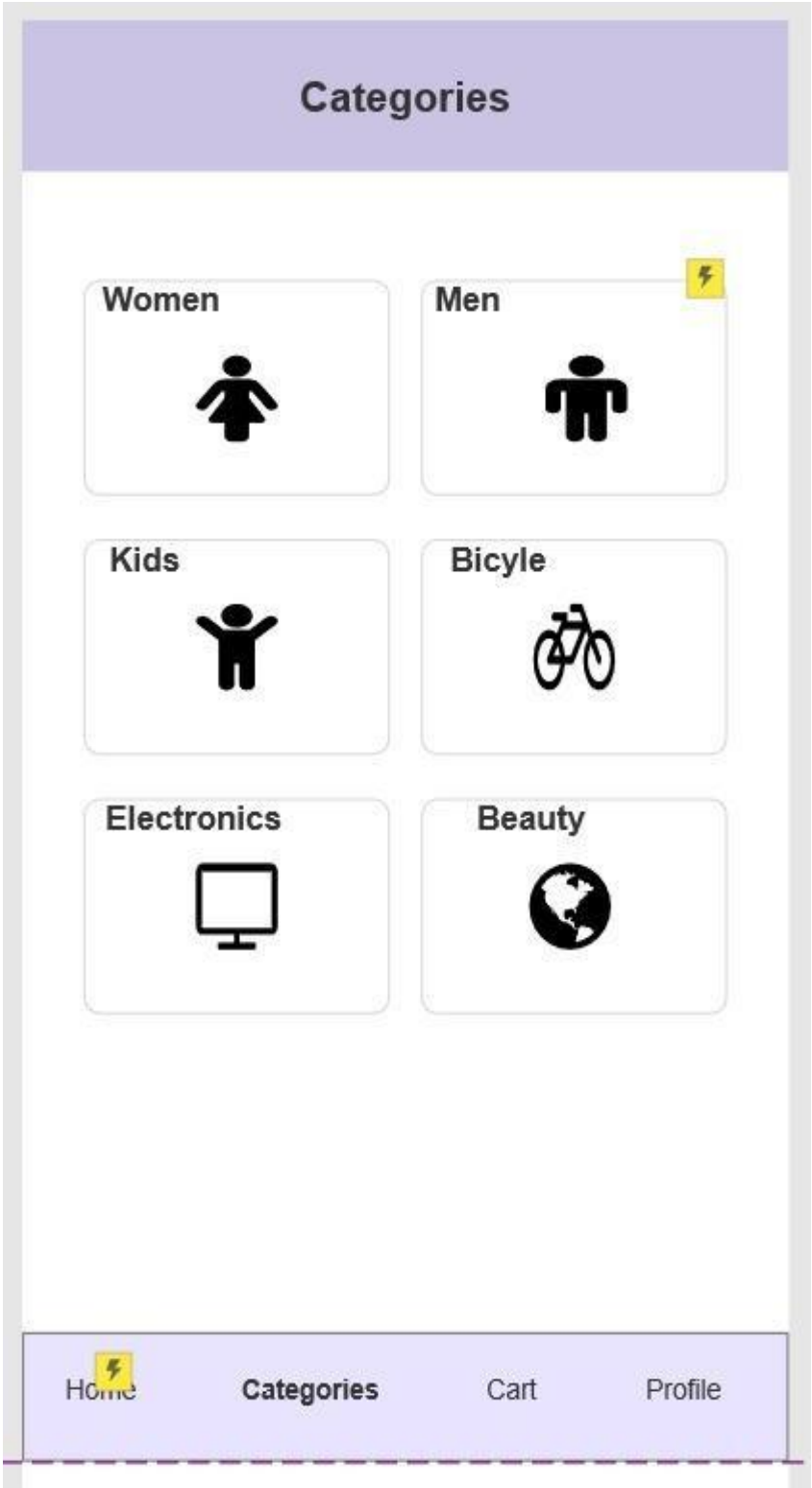
<u>Phase 2</u>: User Design

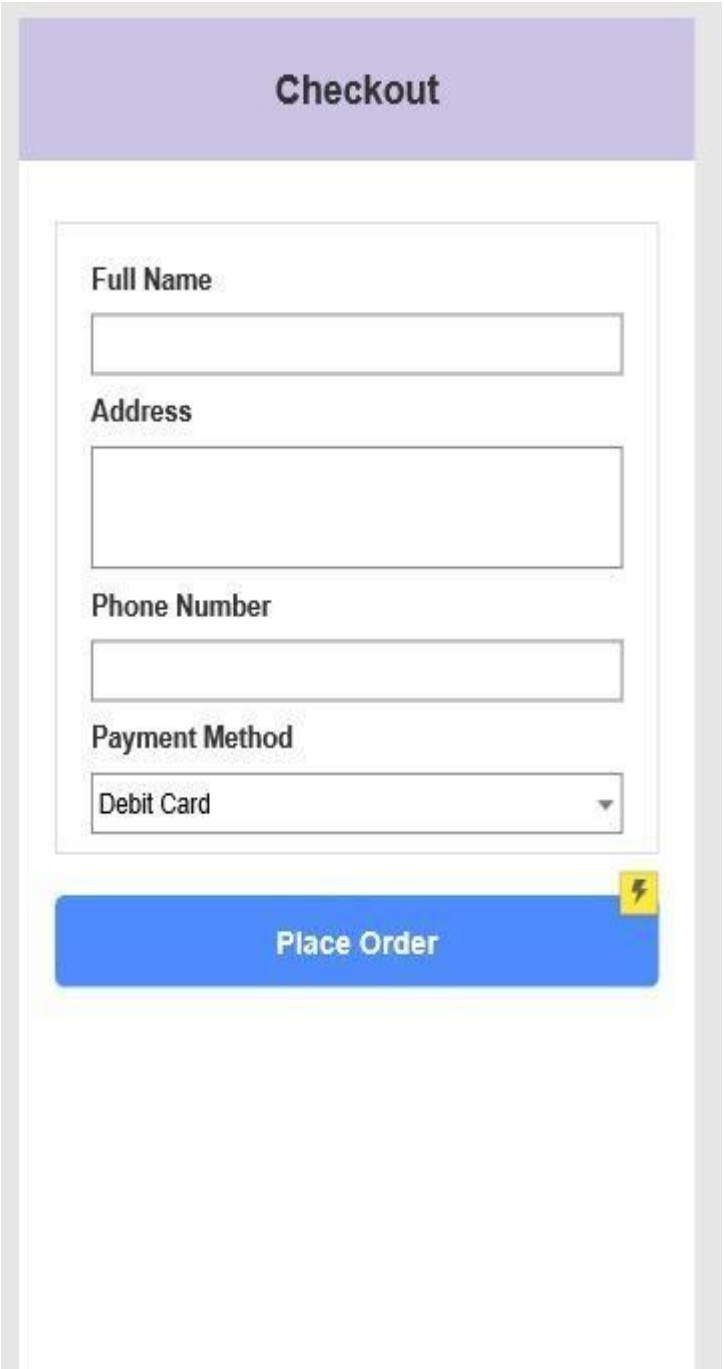
Designers and users collaborate to **create wireframes and prototypes**. The idea is to show how the application will look and behave before any actual coding starts.

1) **Tools Used: Axure**: Used to design the wireframes and UI prototype.
2) **Wireframes Designed:**
   - **Home Page:** Highlights featured products and links to categories. • **Product Categories:** Lists all product categories to explore.
   - **Product Listings:** Shows products under a selected category.
   - **Product Details:** Displays product info with "Add to Cart" option. • **Cart:** Shows added items with price and quantity details.
   - **Checkout:** Collects shipping and payment information. • **Order Confirmation:** Confirms successful order placement.

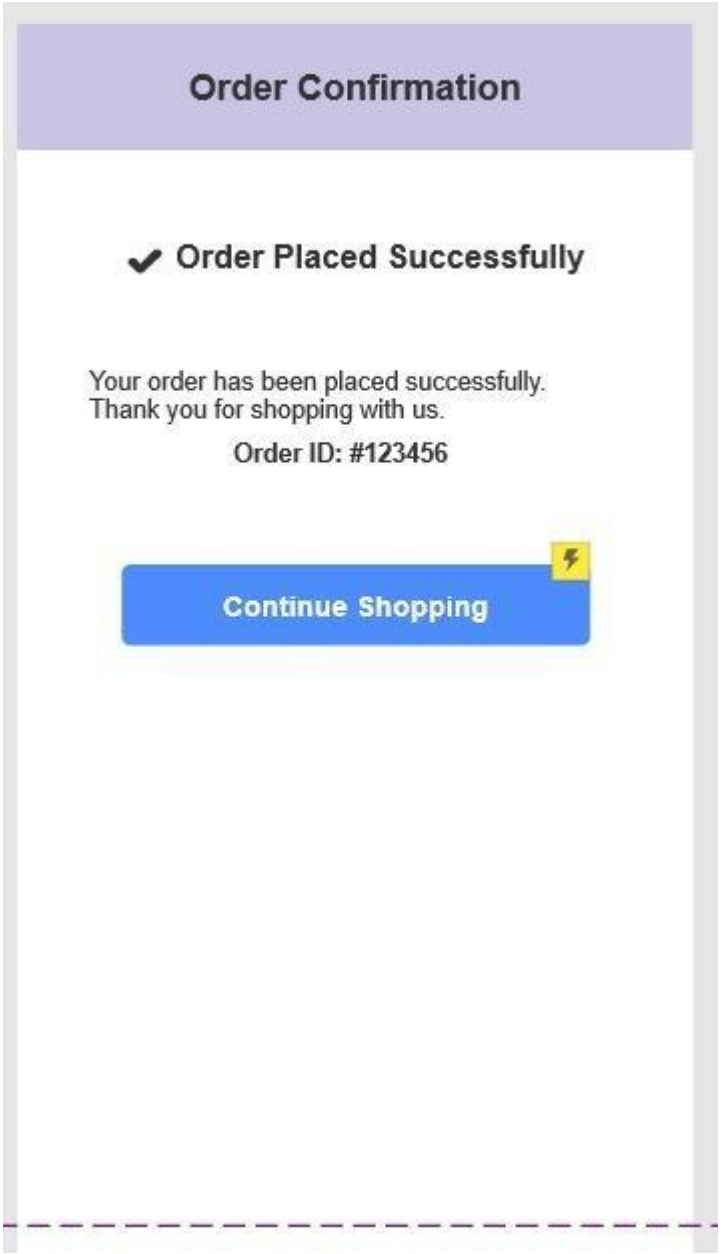
Designed **interactions and navigation** using Axure's Prototype mode Reused components like headers/footers using **Components** Annotated elements for clarity (e.g., what a button does, purpose of sections)

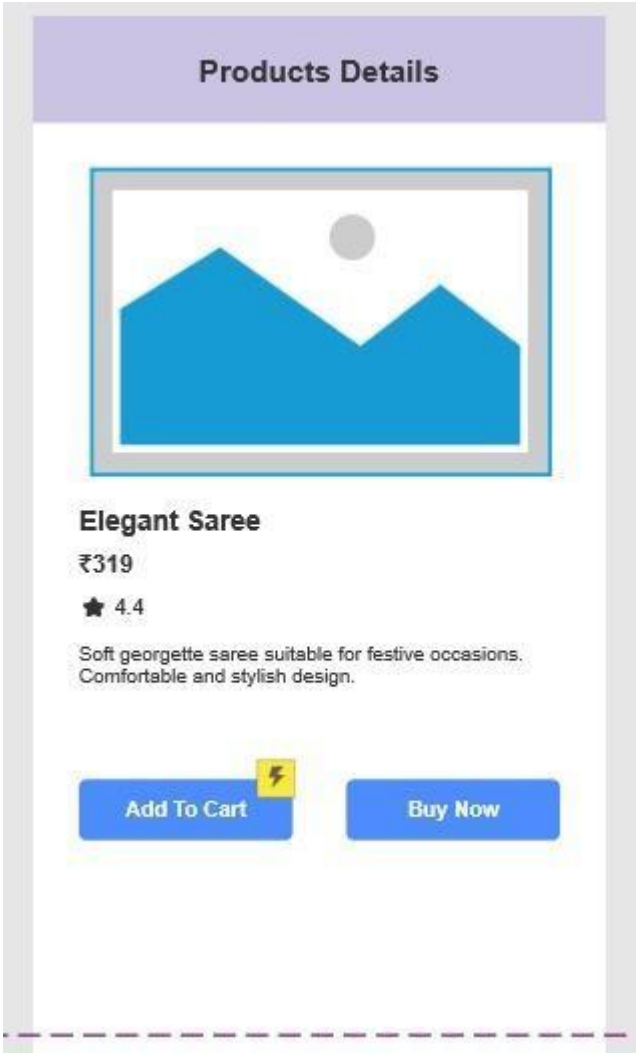

My Cart
Elegant Saree
₹319
Qty: 1
Subtotal: ₹319
Delivery: ₹40
Total: ₹359
Proceed to Checkout
Cart
Profile

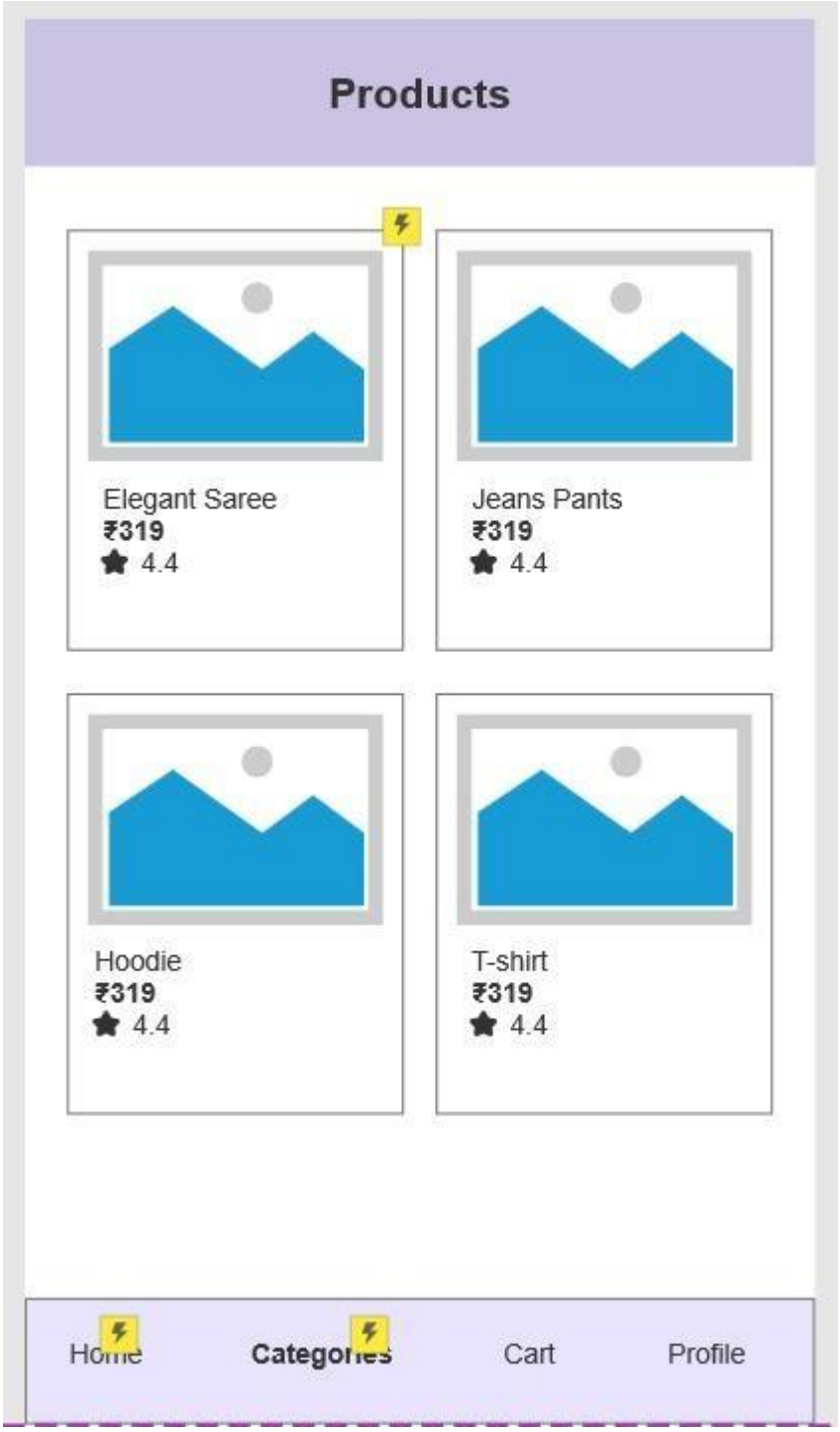

Categories
Women
Men
Kids
Bicyle
Electronics
Beauty
Home
Categories
Cart
Profile


Checkout
Full Name
Address
Phone Number
Payment Method
Debit Card
Place Order

Shopkart
Search for products
Products for you
Elegant Saree
₹319
4.4
Jeans Pants
₹319
4.4
Hoodie
₹319
4.3
T-shirt
₹500
4.0
Home
Cart
Profile


Order Confirmation
Order Placed Successfully
Your order has been placed successfully.
Thank you for shopping with us.
Order ID: #123456
Continue Shopping


Products Details
Elegant Saree
₹319
4.4
Soft georgette saree suitable for festive occasions.
Comfortable and stylish design.
Add To Cart
Buy Now



<u>Phase 3</u>: Construction

This is where the **actual development happens**, but in RAD, it is tightly coupled with ongoing testing and feedback. You refine the prototype based on user responses.

1) Interactive Prototypes:

- Created using Axure's **prototype linking** features.

- Functional paths designed:
    - From **Home → Product Categories → Product Listings → Product Details → Add to Cart → Checkout → Order Confirmation**

2) Dynamic Elements:

- Simulated transitions and modals for adding to cart and checking out. • Screens simulate real-time behavior such as navigating between product listings and cart.

3) Testing and Iteration:

- Prototype previewed with **"Present"** mode in Figma.
- Collected feedback from stakeholders (assumed for documentation). • Iterated based on visual hierarchy, accessibility, and flow suggestions.

<u>Phase 4</u>: Cutover

In this final phase, the system goes live. This includes **finalizing the design**, **training users**, and **preparing documentation** for developers.

1) Finalization and Export:

- Final design prototype shared through Figma link.
- Ready for hand-off to development team.
- Could be exported to **HTML/CSS** using plugins like Figma-to-HTML or further tools if needed.

2) User Training & Support:

- Walkthrough of the prototype for stakeholders or development team. • Notes/documentation provided for developers to understand functionality and user flow.
- Easy access to Figma comments for further collaboration and questions.